DBMS PROJECT REPORT

WEEK 8: DATABASE RELATIONSHIP ANALYSIS USING JOINS

1. AIM

To retrieve and analyse related information from multiple tables in an e-commerce database using SQL JOIN operations.

2. OBJECTIVES

- To understand the use of SQL JOIN operations.
- To combine customer, order, product and payment information.
- To identify customers who have not placed orders.
- To retrieve customer purchase history.
- To generate sales, order and payment reports.

3. INTRODUCTION

SQL JOIN is used to retrieve data from two or more tables based on related columns. In an e-commerce database, customer details, order information, product details and payment transactions are stored in separate tables.

This project uses INNER JOIN, LEFT JOIN and RIGHT JOIN to retrieve related information. Aggregate functions such as COUNT() and SUM() are used to calculate order counts and sales totals. The project also generates business reports to help understand customer purchases and payment transactions.

4. TABLE DESCRIPTIONS

4.1 Customer Table

Stores customer information.

- Customer_ID
- Customer_Name
- Email

- City

4.2 Orders Table

Stores order information.

- Order_ID
- Customer_ID
- Order_Date
- Total_Amount
- Order_Status

4.3 Product Table

Stores product information.

- Product_ID
- Product_Name
- Price
- Category_ID

4.4 Payment Table

Stores payment transaction information.

- Payment_ID
- Order_ID
- Payment_Mode
- Payment_Status

4.5 Order_Details Table

This additional table is used to connect orders with products and their quantities.

- Order_ID
- Product_ID
- Quantity

Note: The Order_Details table is assumed to exist for the product-related queries. Its structure should match the actual database.

5. INNER JOIN

Definition: INNER JOIN returns only the records that match in both tables.

Query 1: Display customer details along with their orders

```
SQL> SELECT C.*,O.* FROM CUSTOMER C INNER JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID;

CUSTOMER_ID FIRST_NAME                                         LAST_NAME
----------- -------------------------------------------------- --------------------------------------------------
EMAIL                                                                                                PHONE           CREATED_A ORDER_I CUSTOMER_ID
---------------------------------------------------------------------------------------------------- --------------- --------- ------- -----------
ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
--------- ------------ --------------------
          1 Arun                                               Kumar
arun@gmail.com                                                                                       9000000001      01-AUG-26 0001            1
28-SEP-26         2800 DELIVERED

          2 Meena                                              Ravi
meena@gmail.com                                                                                      9000000002      02-AUG-26 0002            2
25-SEP-26         1500 SHIPPED

          3 Karthik                                            Raj
karthik@gmail.com                                                                                    9000000003      03-AUG-26 0003            3
04-OCT-26         3000 PENDING

          4 Swetha                                             Mohan
swetha@gmail.com                                                                                     9000000004      04-AUG-26 0004            4
07-OCT-26         4000 DELIVERED

          5 Imran                                              Ali
imran@gmail.com                                                                                      9000000005      05-AUG-26 0005            5
12-OCT-26         1000 DELIVERED

          6 Keerthi                                            Suresh
keerthi@gmail.com                                                                                    9000000006      06-AUG-26 0006            6
18-OCT-26         1200 PENDING

          7 Rohan                                              Sharma
rohan@gmail.com                                                                                      9000000007      07-AUG-26 0007            7
20-OCT-26         5000 SHIPPED

          8 Pooja                                              Nair
pooja@gmail.com                                                                                      9000000008      08-AUG-26 0008            8
25-OCT-26          800 SHIPPED

          9 Sanjay                                             Das
sanjay@gmail.com                                                                                     9000000009      09-AUG-26 0009            9
30-OCT-26          300 DELIVERED

         10 Harini                                             Krishnan
harini@gmail.com                                                                                     9000000010      10-AUG-26 0010           10
01-NOV-26          500 PENDING


10 rows selected.
```

Query 2: Display order details with payment information.

```
SQL> SELECT OD.*,PY.* FROM ORDER_DETAILS OD INNER JOIN PAYMENT PY ON OD.ORDER_ID = PY.ORDER_ID;

ORDER_DETA ORDER_I PRODUCT_ID   QUANTITY      PRICE PAYMENT_ID   ORDER_ID PAYMENT_D PAYMENT_MODE         PAYMENT_STATUS       TRANSACTION_AMOUNT
---------- ------- ---------- ---------- ---------- ---------- ---------- --------- -------------------- -------------------- ------------------
0002       0002    P200002             2         15        102          2 02-OCT-26 Card                 Success                            2500
```

Query 3: Retrieve products purchased by customers.

```
SQL> SELECT P.*,C.*,O.* FROM PRODUCT P JOIN ORDER_DETAILS OD ON P.PRODUCT_ID = OD.PRODUCT_ID JOIN ORDERS O ON OD.ORDER_ID = O.ORDER_ID JOIN CUSTOMER C ON O.CUSTOMER_ID = C.CUSTOMER_ID;

PRODUCT PRODUCT_NAME        CATEGOR      PRICE STOCK_QUANTITY CUSTOMER_ID FIRST_NAME
------- ------------------- ------- ---------- ------------- ----------- --------------------------------------------------
LAST_NAME
--------------------------------------------------
EMAIL                                                                                                PHONE           CREATED_A ORDER_I CUSTOMER_ID
---------------------------------------------------------------------------------------------------- --------------- --------- ------- -----------
ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
--------- ------------ --------------------
P200002 Kids Shoes          1500           15            2          2 Meena
Ravi
meena@gmail.com                                                                                      9000000002      02-AUG-26 0002            2
25-SEP-26         1500 SHIPPED
```

6. LEFT JOIN

Definition: LEFT JOIN returns all records from the left table and matching records from the right table. If no matching record exists, the right-table columns contain NULL.

Query 1: Display all customers, including those who have not placed orders

```
SQL> SELECT C.*,O.* FROM CUSTOMER C LEFT JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID;

CUSTOMER_ID FIRST_NAME                                         LAST_NAME
----------- -------------------------------------------------- --------------------------------------------------
EMAIL                                                                                                PHONE          CREATED_A ORDER_I CUSTOMER_ID
---------------------------------------------------------------------------------------------------- -------------- --------- ------- -----------
ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
--------- ------------ --------------------

          1 Arun                                               Kumar
arun@gmail.com                                                                                       9000000001     01-AUG-26 0001              1
28-SEP-26         2800 DELIVERED

          2 Meena                                              Ravi
meena@gmail.com                                                                                      9000000002     02-AUG-26 0002              2
25-SEP-26         1500 SHIPPED

          3 Karthik                                            Raj
karthik@gmail.com                                                                                    9000000003     03-AUG-26 0003              3
04-OCT-26         3000 PENDING

          4 Swetha                                             Mohan
swetha@gmail.com                                                                                     9000000004     04-AUG-26 0004              4
07-OCT-26         4000 DELIVERED

          5 Imran                                              Ali
imran@gmail.com                                                                                      9000000005     05-AUG-26 0005              5
12-OCT-26         1000 DELIVERED

          6 Keerthi                                            Suresh
keerthi@gmail.com                                                                                    9000000006     06-AUG-26 0006              6
18-OCT-26         1200 PENDING

          7 Rohan                                              Sharma
rohan@gmail.com                                                                                      9000000007     07-AUG-26 0007              7
20-OCT-26         5000 SHIPPED

          8 Pooja                                              Nair
pooja@gmail.com                                                                                      9000000008     08-AUG-26 0008              8
25-OCT-26          800 SHIPPED

          9 Sanjay                                             Das
sanjay@gmail.com                                                                                     9000000009     09-AUG-26 0009              9
30-OCT-26          300 DELIVERED

         10 Harini                                             Krishnan
harini@gmail.com                                                                                     9000000010     10-AUG-26 0010             10
01-NOV-26          500 PENDING

         15 Vishal                                             Singh
vishal@gmail.com                                                                                     9000000015     15-AUG-26


CUSTOMER_ID FIRST_NAME                                         LAST_NAME
----------- -------------------------------------------------- --------------------------------------------------
EMAIL                                                                                                PHONE          CREATED_A ORDER_I CUSTOMER_ID
---------------------------------------------------------------------------------------------------- -------------- --------- ------- -----------
ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
--------- ------------ --------------------

         12 Aparna                                             Joseph
aparna@gmail.com                                                                                     9000000012     12-AUG-26


         14 Lakshmi                                            Devi
lakshmi@gmail.com                                                                                    9000000014     14-AUG-26


         13 Faizal                                             Ahmed
faizal@gmail.com                                                                                     9000000013     13-AUG-26


         11 Naveen                                             Khan
naveen@gmail.com                                                                                     9000000011     11-AUG-26


15 rows selected.
```

Query 2: Find customers without any purchases

```
SQL> SELECT C.*,O.* FROM CUSTOMER C LEFT JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID WHERE O.CUSTOMER_ID IS NULL;

CUSTOMER_ID FIRST_NAME                                         LAST_NAME
----------- -------------------------------------------------- --------------------------------------------------
EMAIL                                                                                                PHONE           CREATED_A ORDER_I CUSTOMER_ID
---------------------------------------------------------------------------------------------------- --------------- --------- ------- -----------
ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
--------- ------------ --------------------
         15 Vishal                                             Singh
vishal@gmail.com                                                                                     9000000015      15-AUG-26


         12 Aparna                                             Joseph
aparna@gmail.com                                                                                     9000000012      12-AUG-26


         14 Lakshmi                                            Devi
lakshmi@gmail.com                                                                                    9000000014      14-AUG-26


         13 Faizal                                             Ahmed
faizal@gmail.com                                                                                     9000000013      13-AUG-26


         11 Naveen                                             Khan
naveen@gmail.com                                                                                     9000000011      11-AUG-26
```

Query 3: Display all products, including products with no sales

```
SQL> SELECT P.*,OD.* FROM PRODUCT P LEFT JOIN ORDER_DETAILS OD ON P.PRODUCT_ID = OD.PRODUCT_ID LEFT JOIN ORDERS O ON O.ORDER_ID = OD.ORDER_ID;

PRODUCT PRODUCT_NAME        CATEGOR      PRICE STOCK_QUANTITY ORDER_DETA ORDER_I PRODUCT_ID   QUANTITY      PRICE
------- ------------------- ------- ---------- -------------- ---------- ------- ---------- ---------- ----------
P200002 Kids Shoes          1500            15              2 0002       0002    P200002             2         15
P000005 Face Wash           C000001        180             40
P000003 Face Cream          C000001        250             50
P100001 Laptop              55000           20              1
P000004 Lipstick            C000002        350             30
```

7. RIGHT JOIN

Definition: RIGHT JOIN returns all records from the right table and matching records from the left table.

Query 1: Display all orders with customer information

```
SQL> SELECT ORDERS.ORDER_ID,PAYMENT.PAYMENT_STATUS FROM ORDERS RIGHT JOIN PAYMENT ON ORDERS.ORDER_ID = PAYMENT.ORDER_ID;

ORDER_I PAYMENT_STATUS
------- --------------------
0001    Success
0002    Success
0003    Failed
0004    Success
0005    Failed
0006    Pending
0007    Success
0008    Success
0009    Failed
0010    Success
        Failed
        Success
        Success
        Success
        Pending

15 rows selected.
```

Query 2: Find orders where customer details are missing

```
SQL> SELECT ORDERS.ORDER_ID,CUSTOMER.FIRST_NAME,CUSTOMER.LAST_NAME FROM ORDERS RIGHT JOIN CUSTOMER ON ORDERS.CUSTOMER_ID = CUSTOMER.CUSTOMER_ID WHERE CUSTOMER.CUSTOMER_ID IS NULL;

no rows selected
```

Query 3: Display all payment records with order details

```
SQL> SELECT PY.PAYMENT_STATUS,OD.ORDER_DETAIL_ID FROM PAYMENT PY RIGHT JOIN ORDER_DETAILS OD ON PY.ORDER_ID = OD.ORDER_ID;

PAYMENT_STATUS       ORDER_DETA
-------------------- ----------
Success              0002
```

8. COMPLETE ORDER DETAILS

Aim: To display customer name, product name, quantity, order date, total amount and payment status in a single report.

```
FIRST_NAME                                         LAST_NAME                                          PRODUCT_NAME          QUANTITY ORDER_DAT
-------------------------------------------------- -------------------------------------------------- -------------------- --------- ---------
TOTAL_AMOUNT PAYMENT_STATUS
------------ --------------------
Meena                                              Ravi                                               Kids Shoes                   2 25-SEP-26
        1500 Success
```

9. CUSTOMER PURCHASE HISTORY

Query 1: Display all products purchased by a customer

```
SQL> SELECT C.FIRST_NAME,C.LAST_NAME,P.PRODUCT_NAME FROM CUSTOMER C JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID JOIN ORDER_DETAILS OD ON O.ORDER_ID = OD.ORDER_ID JOIN P
RODUCT P ON OD.PRODUCT_ID = P.PRODUCT_ID WHERE C.CUSTOMER_ID = '12';

no rows selected
```

Query 2: Calculate the total amount spent by each customer

```
SQL> SELECT C.CUSTOMER_ID,C.FIRST_NAME,C.LAST_NAME,SUM(O.TOTAL_AMOUNT) AS TOTAL_SPENT FROM CUSTOMER C JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID GROUP BY C.CUSTOMER_ID
 ,C.FIRST_NAME,C.LAST_NAME;

CUSTOMER_ID FIRST_NAME                                         LAST_NAME                                          TOTAL_SPENT
----------- -------------------------------------------------- -------------------------------------------------- -----------
          1 Arun                                               Kumar                                                     2800
          3 Karthik                                            Raj                                                       3000
          8 Pooja                                              Nair                                                       800
         10 Harini                                             Krishnan                                                   500
          6 Keerthi                                            Suresh                                                    1200
          9 Sanjay                                             Das                                                        300
          4 Swetha                                             Mohan                                                     4000
          5 Imran                                              Ali                                                       1000
          7 Rohan                                              Sharma                                                    5000
          2 Meena                                              Ravi                                                      1500

10 rows selected.
```

Query 3: Count the number of orders placed by each customer

```
SELECT C.CUSTOMER_ID,C.FIRST_NAME,C.LAST_NAME COUNT(O.ORDER_ID) AS TOTAL_ORDERS FROM CUSTOMER C LEFT JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID GROUP BY C.CUSTOMER_ID,
C.FIRST_NAME,C.LAST_NAME
```

```
CUSTOMER_ID FIRST_NAME                                         LAST_NAME                                          TOTAL_ORDERS
----------- -------------------------------------------------- -------------------------------------------------- ------------
          1 Arun                                               Kumar                                                         1
          3 Karthik                                            Raj                                                           1
         13 Faizal                                             Ahmed                                                         0
         11 Naveen                                             Khan                                                          0
          8 Pooja                                              Nair                                                          1
         10 Harini                                             Krishnan                                                      1
          6 Keerthi                                            Suresh                                                        1
          9 Sanjay                                             Das                                                           1
         15 Vishal                                             Singh                                                         0
          4 Swetha                                             Mohan                                                         1
          5 Imran                                              Ali                                                           1
          7 Rohan                                              Sharma                                                        1
         14 Lakshmi                                            Devi                                                          0
          2 Meena                                              Ravi                                                          1
         12 Aparna                                             Joseph                                                        0

15 rows selected.
```

Query 4: Display the latest purchase details of each customer

```
SQL> SELECT C.FIRST_NAME,C.LAST_NAME,O.ORDER_ID,O.ORDER_DATE,O.TOTAL_AMOUNT FROM CUSTOMER C JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID WHERE O.ORDER_DATE = (SELECT MAX
(O2.ORDER_DATE)FROM ORDERS O2 WHERE ORDERS O2.CUSTOMER_ID = C.CUSTOMER_ID);
```

```
FIRST_NAME                                         LAST_NAME                                          ORDER_I ORDER_DAT TOTAL_AMOUNT
-------------------------------------------------- -------------------------------------------------- ------- --------- ------------
Arun                                               Kumar                                              0001    28-SEP-26         2800
Meena                                              Ravi                                               0002    25-SEP-26         1500
Karthik                                            Raj                                                0003    04-OCT-26         3000
Swetha                                             Mohan                                              0004    07-OCT-26         4000
Imran                                              Ali                                                0005    12-OCT-26         1000
Keerthi                                            Suresh                                             0006    18-OCT-26         1200
Rohan                                              Sharma                                             0007    20-OCT-26         5000
Pooja                                              Nair                                               0008    25-OCT-26          800
Sanjay                                             Das                                                0009    30-OCT-26          300
Harini                                             Krishnan                                           0010    01-NOV-26          500

10 rows selected.
```

10. MULTI-TABLE BUSINESS REPORTS

Report 1: Customer Order Report

Displays customer name, order ID, order date and order status.

```
SQL> SELECT C.FIRST_NAME,C.LAST_NAME,O.ORDER_ID,O.ORDER_DATE,O.ORDER_STATUS FROM CUSTOMER C INNER JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID;

FIRST_NAME                                         LAST_NAME                                          ORDER_I ORDER_DAT ORDER_STATUS
-------------------------------------------------- -------------------------------------------------- ------- --------- --------------------
Arun                                               Kumar                                              0001    28-SEP-26 DELIVERED
Meena                                              Ravi                                               0002    25-SEP-26 SHIPPED
Karthik                                            Raj                                                0003    04-OCT-26 PENDING
Swetha                                             Mohan                                              0004    07-OCT-26 DELIVERED
Imran                                              Ali                                                0005    12-OCT-26 DELIVERED
Keerthi                                            Suresh                                             0006    18-OCT-26 PENDING
Rohan                                              Sharma                                             0007    20-OCT-26 SHIPPED
Pooja                                              Nair                                               0008    25-OCT-26 SHIPPED
Sanjay                                             Das                                                0009    30-OCT-26 DELIVERED
Harini                                             Krishnan                                           0010    01-NOV-26 PENDING

10 rows selected.
```

Report 2: Sales Report

Displays product name, quantity sold and total revenue.

```
SQL> SELECT P.PRODUCT_NAME,SUM(OD.QUANTITY)AS QUANTITY_SOLD,SUM(OD.QUANTITY*P.PRICE)AS TOTAL_REVENUE FROM PRODUCT P RIGHT JOIN ORDER_DETAILS OD ON P.PRODUCT_ID = OD.PRODUC
T_ID GROUP BY P.PRODUCT_ID,P.PRODUCT_NAME;

PRODUCT_NAME         QUANTITY_SOLD TOTAL_REVENUE
-------------------- ------------- -------------
Kids Shoes                       2            30
```

Report 3: Payment Analysis Report

Displays payment mode, total transactions and successful payments.

```
SQL> SELECT PAYMENT_MODE,COUNT(*)AS TOTAL_TRANSACTIONS,SUM(CASE WHEN PAYMENT_STATUS = 'SUCCESS' THEN 1 ELSE 0 END)AS SUCCESSFUL_PAYMENTS FROM PAYMENT GROUP BY PAYMENT_MODE
;

PAYMENT_MODE         TOTAL_TRANSACTIONS SUCCESSFUL_PAYMENTS
-------------------- ------------------ ------------------
Net Banking                          3                  0
Card                                 5                  0
Cash                                 2                  0
UPI                                  5                  0
```

Report 4
(a):
Top Purchasing Customers

Displays customers ordered by total spending, from highest to lowest.

```
SQL> SELECT C.FIRST_NAME,C.LAST_NAME,SUM(O.TOTAL_AMOUNT)AS TOTAL_SPENT FROM CUSTOMER C INNER JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID GROUP BY C.CUSTOMER_ID,C.FIRST_
NAME,C.LAST_NAME ORDER BY TOTAL_SPENT DESC;

FIRST_NAME                                         LAST_NAME                                          TOTAL_SPENT
-------------------------------------------------- -------------------------------------------------- -----------
Rohan                                              Sharma                                                    5000
Swetha                                             Mohan                                                     4000
Karthik                                            Raj                                                       3000
Arun                                               Kumar                                                     2800
Meena                                              Ravi                                                      1500
Keerthi                                            Suresh                                                    1200
Imran                                              Ali                                                       1000
Pooja                                              Nair                                                       800
Harini                                             Krishnan                                                   500
Sanjay                                             Das                                                        300

10 rows selected.
```

Report 4(b): Customers with the Maximum Number of Orders

```
SQL> SELECT C.FIRST_NAME,C.LAST_NAME,COUNT(O.ORDER_ID)AS TOTAL_ORDERS FROM CUSTOMER C,ORDERS O WHERE C.CUSTOMER_ID = O.CUSTOMER_ID GROUP BY C.FIRST_NAME,C.LAST_NAME;

FIRST_NAME                                         LAST_NAME                                          TOTAL_ORDERS
-------------------------------------------------- -------------------------------------------------- ------------
Swetha                                             Mohan                                                         1
Arun                                               Kumar                                                         1
Sanjay                                             Das                                                           1
Harini                                             Krishnan                                                      1
Karthik                                            Raj                                                           1
Rohan                                              Sharma                                                        1
Meena                                              Ravi                                                          1
Keerthi                                            Suresh                                                        1
Pooja                                              Nair                                                          1
Imran                                              Ali                                                           1

10 rows selected.
```

Report 4(c): Customers with the Highest Spending

```
SQL> SELECT C.FIRST_NAME,C.LAST_NAME,SUM(O.TOTAL_AMOUNT)AS TOTAL_SPENT FROM CUSTOMER C,ORDERS O WHERE C.CUSTOMER_ID = O.CUSTOMER_ID GROUP BY C.FIRST_NAME,C.LAST_NAME HAVIN
G SUM(O.TOTAL_AMOUNT) = (SELECT MAX(TOTAL_AMOUNT) FROM ORDERS);

FIRST_NAME                                         LAST_NAME                                          TOTAL_SPENT
-------------------------------------------------- -------------------------------------------------- -----------
Rohan                                              Sharma                                                    5000
```

11. EXPECTED RESULTS

After executing the queries with valid data, the results should demonstrate the following:

1. INNER JOIN displays matching customer, order, product and payment information.

2. LEFT JOIN displays all customers or products, including those without matching records.
3. RIGHT JOIN displays all orders or payment records from the right-side table.
4. The complete order report combines the requested order details.
5. Customer purchase history shows purchased products, spending, order counts and latest purchases.
6. Business reports summarise sales, payments and customer spending.

The actual output depends on the records stored in the database. The queries should be executed before inserting screenshots or actual result tables into the final submission.

12. CONCLUSION

This project demonstrates how SQL JOIN operations combine related data from multiple tables in an e-commerce database. INNER JOIN retrieves matching records, LEFT JOIN includes all records from the left table, and RIGHT JOIN includes all records from the right table. Aggregate functions help calculate sales, spending and order counts. Thus, SQL JOINs are useful for retrieving information and generating business reports from relational databases.